

Ayan Majumdar

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About

Doctoral Researcher at MPI-SWS specializing in Trustworthy AI. I am passionate about bridging the gap between theoretical research in fairness, causality, and explainability and the practical deployment of safety-compliant Automated Decision-Making and Generative AI systems.

Research Interests

Trustworthy AI 📖 Responsible AI, AI Safety, Fairness, Explainability, Causality

Deep Learning 📖 Generative AI, Large Language Models, Vision Language Models, Model Alignment

Education

2021 – 2026 (expected)

📖 Ph.D. in Computer Science

MPI-SWS and Saarland University, Saarbrücken, Germany

Area: *Deep Learning, Generative AI, Fairness, Explainability*

Advisor: Prof. Dr. Krishna P. Gummadi, Prof. Dr. Isabel Valera

Contact: [gummadi\[at\]mpi-sws.org](mailto:gummadi[at]mpi-sws.org), [ivalera\[at\]cs.uni-saarland.de](mailto:ivalera[at]cs.uni-saarland.de)

2017 – 2021

📖 M.Sc. in Computer Science

Saarland University, Saarbrücken, Germany

Thesis: *Generating Counterfactuals for Causal Fairness*

Outline: *Deep generative models and their implicit assumptions in generating counterfactuals from observed data in the context of fairness.*

Advisor: Prof. Dr. Krishna P. Gummadi, Prof. Dr. Isabel Valera

GPA: 1.2/1.0 (German Scale)

2011 – 2015

📖 B.Tech. in Electronics & Communication

Heritage Institute of Technology, Kolkata, India

Project: *Automated traffic detection using image processing*

Outline: *Utilize blob detection techniques for detecting traffic from video sequences.*

Supervisor: Prof. Anindya Sen

GPA: 8.8/10.0

Publications

Conference Proceedings

- 1 **Majumdar, A.**, & Valera, I. (2024). CARMA: A practical framework to generate recommendations for causal algorithmic recourse at scale. In *2024 ACM Conference on Fairness, Accountability, and Transparency* (pp. 1745–1762). 📄 Paper 🔄 Code.
- 2 Nanda, V., **Majumdar, A.**, Kolling, C., Dickerson, J. P., Gummadi, K. P., Love, B. C., & Weller, A. (2023). Do invariances in deep neural networks align with human perception? In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 37, pp. 9277–9285). 📄 Paper 🔄 Code.
- 3 Rateike*, M., **Majumdar*, A.**, Mineeva, O., Gummadi, K. P., & Valera, I. (2022). Dont Throw it Away! The Utility of Unlabeled Data in Fair Decision Making. In *2022 ACM Conference on Fairness, Accountability, and Transparency* (pp. 1421–1433). 📄 Paper 🔄 Code.

Archival Preprints

- 4 **Majumdar*, A.**, Kanubala*, D. D., Gupta, K., & Valera, I. (2026). A causal framework to measure and mitigate non-binary treatment discrimination. *arXiv preprint arXiv:2503.22454*. Accepted to AAAI 2026  Paper  Code.
- 5 **Majumdar, A.**, Chen, F., Li, J., & Wang, X. (2025). Evaluating llms for demographic-targeted social bias detection: A comprehensive benchmark study. *arXiv preprint arXiv:2510.04641*.  Paper.

Work Experience

- Aug. 2024 – Feb. 2025  **Ph.D. Research Intern**
Huawei Research TTE-DE Lab, München, Germany
Project: Scaling-up Data Bias Detection & Survey on AI Governance
Role: Developed a benchmark using LLMs to detect demographic-targeted social bias in large datasets (preprint [5]). Conducted research surveys to support AI governance and alignment with regulations like the EU AI Act.
- Apr. 2021 – ongoing  **Doctoral Researcher**
Max Planck Institute for Software Systems, Saarbrücken, Germany
Role: Leading multiple projects on improving the practicality of Responsible AI and Safety in the context of Automated Decision-Making and Generative AI.
Supervisor: Prof. Dr. Krishna P. Gummadi, Prof. Dr. Isabel Valera
- Oct. 2019 – Mar. 2021  **Research Assistant**
Max Planck Institute for Software Systems, Saarbrücken, Germany
Project: *Exploring bias and fairness with deep generative models*
Role: Led project regarding exploration of bias in deep generative models for facial image data; designed core methodologies and evaluations.
Supervisor: Prof. Dr. Krishna P. Gummadi
- Apr. 2018 – Mar. 2019  **Research Assistant**
SFB1102, Saarland University, Saarbrücken, Germany
Project: *Mutual Intelligibility in Slavic Languages*
Role: Engineered automated data processing pipelines to collect and clean large-scale textual datasets for Machine Translation experiments. Developed web-based user studies to evaluate mutual intelligibility in Slavic languages.
Supervisor: Prof. Dr. Dietrich Klakow
- Jul. 2015 – Aug. 2017  **Systems Engineer**
Infosys Ltd., Bengaluru, India
Role: Optimized and validated Session Initiation Protocol (SIP) and VoIP functionalities within session border controllers. Recognized with the Infy Insta Award (2017) and Spot Award (2016) for critical project contributions.

Technical Skills

Generative AI & LLM Engineering

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| Frameworks |  Transformers, vLLM, Sentence-Transformers, FAISS |
| Techniques |  Prompt Optimization, In-Context Learning, Multi-modal Inference, RAG, Fine-tuning, Benchmarking, LLM-as-a-Judge, Automated Content Moderation |

Responsible AI & AI Safety

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| Frameworks |  AIF360, Fairlearn, CleverHans, Foolbox, robustness |
| Techniques |  Causal Inference, Counterfactual Explanations, Statistical & Causal Fairness |

Technical Skills (continued)

Programming & Core Infrastructure

Core ML	PyTorch, Lightning, Scikit-learn, NumPy, Pandas, Scipy, Optuna
Languages & Tools	Python, Git, L ^A T _E X, Weights & Biases, Matplotlib, Seaborn

Teaching Assistance

Summer 2024	Machine Learning (Core Lecture) , Saarland University
Summer 2021	Machine-Assisted Decision Making (Seminar) , Saarland University
Summer 2019	Statistical Natural Language Processing (Advanced Lecture) , Saarland University

Mentoring Experience

May 2024 – July 2024	Co-mentored Pranav Nyati (B.Tech. student, IIT-Kharagpur) during DAAD summer internship on algorithmic recourse using swarm algorithms .
Sept. 2025 – Jan. 2026	Co-mentored Lena Marie Budde (M.Sc. student, Saarland University) during thesis and project on personalization in algorithmic recourse .

Talks and Posters

2024	Scaling Up Causal Algorithmic Recourse with CARMA <i>European Workshop on Algorithmic Fairness 2024, Center for Perspicuous Computing 2024 Meeting</i>
2022	Don't Throw it Away! The Utility of Unlabeled Data in Fair Decision Making <i>SAP Innovation Sessions, MPI for Intelligent Systems, MILA Quebec AI Institute</i>

Online Certifications

Coursera	Algorithmic toolbox, Data structures, Graph algorithms, String algorithms
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Academic Activities

Program Committee	FAccT 2024-25, ICLR 2023-25, ICML 2023-24, NeurIPS 2023/25, WebConf-Web4Good 2026, AAAI 2025-26, AIES 2025
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Language Skills

Native/Fluent	English	●●●●●●	Bengali	●●●●●●
Proficient/Intermediate	German	●●●●●●	Hindi	●●●●●●

Achievements & Awards

- Infy Insta award (2017) and Spot award (2016) for significant contributions to the project, Infosys, India.